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Exploring the evolution of artificial intelligence in education: from AI-guided learning to learner-personalized paradigms

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ABSTRACT

Advancements in computing and information technologies have significantly transformed educational practices. This transformation can be understood through three paradigms. The first paradigm (Artificial Intelligence Guided Learning) involves systems that model knowledge and guide cognitive processes, positioning learners as passive recipients. The second paradigm (Artificial Intelligence Interactive Learning) fosters collaboration between learners and intelligent systems, enabling interactive learning experiences. The third and most advanced paradigm (Artificial Intelligence Driven Personalized Learning) places learners at the center, using intelligent systems to personalize education and empower students to take charge of their learning journey. These paradigms illustrate a trajectory toward learner-centric approaches that emphasize personalization, adaptability, and agency. Intelligent systems in education increasingly support reflection and iterative learning, where students actively shape their educational experiences, and the systems evolve based on real-time learner data. This transformation promises to make education more flexible, responsive, and tailored, addressing diverse learning needs while advancing a data-driven, personalized approach to pedagogy.

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SUBJECTS

Artificial Intelligence;
Technology in Education;
Educational Technology

Introduction

The ongoing evolution of computing and information processing has fundamentally transformed education through the integration of artificial intelligence (AI) (Forero-Corba & Bennasar, 2023; Namatovu & Kyambade, 2025). Artificial Intelligence in Education (AIED) has expanded possibilities for teaching and learning by introducing applications such as intelligent tutoring systems (ITS), adaptive learning platforms, learning analytics dashboards, and advanced human-computer interactions. These tools enable personalization, enhance engagement, and provide tailored support for diverse learner needs. Over the past three decades, AIED has emerged as a transformative force, driving new paradigms in instructional design and research (Ouyang & Jiao, 2021).

AIED's advancements leverage cutting-edge technologies such as machine learning, deep learning, and natural language processing to create intelligent learning environments capable of analyzing learner behaviors, building predictive models, and offering adaptive recommendations (Ouyang & Jiao, 2021; Rowe & Baepler, 2021). Beyond technological innovations, AIED has opened new avenues for personalized learning, shifting traditional classroom-based education toward learner-centered models. It has the potential to foster agency, improve lifelong learning outcomes, and enhance educational equity by tailoring experiences to individual needs (Pinkwart, 2016; Qin & Zhang, 2025).

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However, AI's integration into education presents significant challenges. Research underscores that successful educational outcomes require more than advanced technology; they depend on aligning AI tools with robust pedagogical principles to address the complexities of teaching and learning (Kyambade, Namatovu et al., 2025). Despite the extensive literature on AIED approaches, categorizations, and challenges, gaps persist in understanding the interplay between AI technologies, existing educational theories, and their practical implications for instructional quality (Hwang et al., 2020).

Recent systematic reviews have provided valuable insights into the evolving landscape of AI applications in education. For instance, Hardaker and Glenn (2025) and Namatovu and Kyambade (2025) conducted a comprehensive review highlighting how AI technologies facilitate personalized learning by adapting instruction to individual learner needs, thereby enhancing engagement and academic outcomes. Similarly, Liu et al. (2025) examined the role of ITS and Robot Tutoring Systems (RTS) in delivering personalized and adaptive instruction, emphasizing their potential to bridge learning gaps and improve student proficiency in core academic areas.

Moreover, the integration of AI in educational assessment has been explored in recent literature. A systematic review by Maity and Deroy (2024), analyzed the impact of AI on assessment methods in primary and secondary education, revealing that AI tools can predict student performance, automate evaluations, and provide more objective assessments through technologies like neural networks and natural language processing. These findings underscore the transformative potential of AI in enhancing assessment practices across educational levels.

Ethical considerations and the human-centered design of AI systems in education have also been the focus of recent studies. Alfredo et al. (2024) conducted a systematic literature review emphasizing the importance of involving educational stakeholders in the design process of AI and Learning Analytics systems to ensure trustworthiness, reliability, and alignment with pedagogical goals. Additionally, Yan et al. (2024) explored the practical and ethical challenges of implementing large language models in educational contexts, highlighting concerns such as data privacy, transparency, and the need for human oversight in AI-driven educational tools.

This article seeks to bridge these gaps by introducing three paradigms of AIED. Each paradigm reflects a distinct approach to integrating AI into education, illustrating how these technologies interact with learning theories and influence instructional strategies. By exploring theoretical, pedagogical, and computational dimensions, this framework provides insights into how AIED can empower learners, promote equitable access to education, and prepare institutions for the demands of the knowledge-driven age. The discussion aims to guide future research and practice, emphasizing the importance of leveraging AI to create transformative, learner-centered educational ecosystems.

Literature review

Artificial intelligence in education (AIED)

Artificial intelligence in education (AIED) continues to grapple with foundational challenges, such as how to cater to learners' diverse needs, provide timely and relevant support, and empower learners to take active control of their educational journeys (Du Boulay, 2021; Holmes et al., 2019). While AIED leverages advanced computational techniques, the successful implementation of these technologies does not inherently lead to improved learning outcomes (Ouyang & Jiao, 2021). Effective integration of AI tools requires alignment with established educational theories to inform design and practice (Bower, 2021; Luckin et al., 2016).

Research reviews consistently highlight the limited connection between AI methodologies and theoretical frameworks, which often diminishes the impact of AIED implementations (Kim & Kwon, 2025; Kyambade et al., 2024). Zawacki-Richter et al. (2019), in an analysis of 146 studies on AI in higher education, observed a significant gap in addressing the pedagogical and ethical implications of AI systems. Similarly, Chen et al. (2020) reviewed 45 influential studies and found that only a few explicitly referenced foundational theories, such as situated learning and collaborative learning, to guide their development. Another systematic review by Deeva et al. (2021) analyzing 109 studies on automated feedback

tools revealed that most lacked theoretical grounding, even though theories are crucial for contextualizing how these systems are utilized.

Additionally, the philosophical perspectives underpinning AI tools influence how learners engage with these systems. For instance, the lack of theoretical grounding may inadvertently position AI as merely a technical solution rather than a meaningful pedagogical partner. This limitation is particularly critical, given that the success of educational technologies often hinges on their alignment with learner-centered paradigms and adaptive teaching strategies (Hwang et al., 2020).

In light of these challenges, this article synthesizes the evolution of AIED paradigms by categorizing the roles of AI within theoretical, pedagogical, and practical contexts. It proposes a framework to bridge existing gaps by integrating theoretical underpinnings into AIED development, thereby fostering personalized, learner-driven education and guiding future research and applications in the field.

Benefits of AIED processes

Artificial intelligence (AI) has increasingly become an integral component of educational systems worldwide, offering transformative potential in teaching and learning processes. AI technologies provide personalized learning experiences by adapting to individual student needs, thereby enhancing engagement and academic performance (Ward et al., 2024). For instance, adaptive learning systems utilize AI algorithms to tailor educational content to students' unique learning styles and paces, leading to improved comprehension and retention (Wikipedia, 2024). Additionally, AI streamlines administrative tasks such as grading and scheduling, allowing educators to allocate more time to instruction and student support (Lu, 2025). Automated grading systems can efficiently assess multiple-choice tests and short-answer questions, providing immediate feedback to students and reducing educators' workload (Medium, 2024).

AI also enhances accessibility and inclusivity in education. Tools like speech-to-text, text-to-speech, and real-time captioning assist students with disabilities, while AI-powered translation services help non-native speakers comprehend educational materials, fostering a more inclusive learning environment (Ouyang & Jiao, 2021; TeachEducator, 2024). Furthermore, AI-powered ITS offer individualized support to students outside the traditional classroom setting. These systems can simulate one-on-one tutoring by providing explanations, answering questions, and offering feedback, which is particularly beneficial for students requiring additional assistance (Bigscal Technologies, 2024). In addition, AI facilitates data-driven decision-making in education. By analyzing vast amounts of educational data, AI can generate insights into student performance, helping educators identify learning gaps and tailor instructional strategies accordingly (Informal Pakistan, 2024). Lastly, AI contributes to the creation of immersive and interactive learning environments. Through gamification and virtual reality, AI can make learning more engaging, thereby increasing student motivation and participation (Tech AI, 2024).

Challenges of AI application in education

Despite these benefits, the integration of AI in education presents several challenges. Data privacy and security are paramount concerns, as AI systems collect and analyze sensitive student information, raising issues about confidentiality and potential misuse (Kyambade et al., 2023; Liu, 2024). Moreover, AI algorithms trained on biased data can perpetuate existing inequalities, affecting fair access to educational opportunities (Rivera, 2024). Over-reliance on AI may also diminish essential human elements of teaching, such as empathy and mentorship, which are crucial for holistic educational development (White, 2024). Furthermore, the financial investment required for AI infrastructure and training can be prohibitive, especially for underfunded institutions, potentially exacerbating the digital divide (Bates, 2023).

AI systems in education often require the collection and analysis of vast amounts of student data, raising significant concerns about privacy and security. Without robust data protection measures, there's a heightened risk of unauthorized access, data breaches, and misuse of personal information. Khan (2025) emphasizes that educational institutions must establish stringent data governance frameworks to safeguard student information and maintain trust among stakeholders.

AI algorithms are only as unbiased as the data they are trained on. If the training data contains historical biases, AI systems can inadvertently perpetuate and even exacerbate existing inequalities within the

educational system. Zhang (2024) notes that biased algorithms can lead to discriminatory outcomes, such as unfair grading or unequal access to learning resources, thereby reinforcing educational disparities. The increasing reliance on AI tools in classrooms may lead to a reduction in human interaction, which is vital for the development of social and emotional skills among students. AI lacks the capacity for empathy, nuanced understanding, and the ability to respond to the emotional needs of students—qualities intrinsic to human educators. Novianti (2025) argues that overemphasis on AI-driven instruction could hinder the cultivation of critical soft skills such as communication, collaboration, and emotional intelligence.

Implementing AI technologies in educational institutions requires substantial financial investment in hardware, software, and ongoing maintenance. Many schools, particularly those in underfunded or rural areas, may struggle to afford these costs, leading to a digital divide where only well-resourced institutions can benefit from AI advancements. UNESCO (2023) highlights that such disparities can exacerbate existing educational inequalities, making it imperative to develop inclusive strategies that ensure equitable access to AI tools. Effective integration of AI into the classroom hinges on educators' ability to utilize these tools proficiently. However, many teachers lack the necessary training and support to implement AI technologies effectively. Holstein et al. (2020) emphasize that without proper professional development, educators may resist adopting AI tools or use them ineffectively, thereby diminishing their potential benefits.

The use of AI in education also brings forth ethical dilemmas, including concerns about academic integrity. Students might misuse AI tools to generate assignments or answers, undermining the authenticity of their work. Khan (2025) discusses the ethical implications of surveillance through AI-powered proctoring systems, which can infringe on student privacy and create a conflict between maintaining academic integrity and respecting individual rights.

Recommendations for improving AI application in educational programs

To maximize the benefits and mitigate the challenges of AI in education, several strategies are recommended. Developing comprehensive ethical guidelines is essential to govern data usage, privacy, and ethical considerations in AI applications. Porayska-Pomsta et al. (2023) and Kyambade and Namatovu (2025) emphasize the importance of integrating ethical principles into the design and implementation of AI systems to ensure they support educational values and respect learners' rights. Similarly, the European Commission (2023) has outlined ethical guidelines focusing on transparency, accountability, and fairness in AI usage within educational settings. The adoption of Explainable AI (XAI) systems ensures transparency, allowing educators and students to understand how decisions are made. Gillani et al. (2023) highlight that XAI can enhance trust in AI applications by making their operations more transparent and understandable. Altukhi and Pradhan (2025) further argue that explainability is crucial for educators to effectively integrate AI tools into their teaching practices, as it enables them to comprehend and trust the AI's recommendations. Investing in professional development provides ongoing training for educators to effectively integrate AI tools into their teaching practices (Al Mansoor, 2023). Promoting inclusive access by addressing the digital divide ensures all students have access to necessary technology and resources, fostering equity in educational opportunities (UNESCO, 2023).

Providing ongoing training for educators is vital for the effective integration of AI tools into teaching practices. Al Mansoor (2023) and Kyambade, Namatovu et al. (2025) underscores the need for professional development programs that equip teachers with the necessary skills to utilize AI technologies effectively. Such training ensures that educators can leverage AI tools to enhance learning outcomes while addressing any ethical or practical concerns that may arise. Ensuring equitable access to AI technologies is crucial to prevent exacerbating existing educational inequalities. UNESCO (2023) and Kyambade, Tushabe et al. (2025) advocates for policies that promote inclusive access to AI tools, particularly for students in under-resourced communities. This includes investing in infrastructure, providing affordable access to technology, and developing AI applications that are culturally and linguistically appropriate. Engaging various stakeholders, including educators, students, policymakers, and technologists, in the development and implementation of AI in education fosters systems that are more responsive to the needs of all users. Porayska-Pomsta et al. (2023) emphasize the importance of participatory

design approaches that involve stakeholders in the creation of AI tools, ensuring that these technologies align with educational goals and ethical standards.

Methodology

This study aims to synthesize major paradigms regarding the roles of AI in education, connecting them to established educational theories and exploring their practical implications. The research is driven by three fundamental questions:

- What roles does AI play in education?
- How does AI relate to existing educational theories?
- To what extent do AI technologies influence learning and instruction?

To answer these questions, a systematic literature review was conducted following the PRISMA guidelines. The review process encompassed the following steps:

1. **Data Sources:** The research utilized a range of academic databases, including Web of Science, Taylor & Francis, Scopus, ScienceDirect, Wiley Online Library, IEEE, ACM, and EBSCO. This comprehensive approach ensured the inclusion of diverse perspectives and findings from various educational and technological disciplines.
2. **Search Strategy:** The search strategy employed Boolean operators and truncation symbols to capture a broad spectrum of relevant literature. The search terms were applied to the title, abstract, and keywords fields. [Table 1](#) outlines the search formula:
3. **Time Frame:** The review primarily covered articles published from 2000 to 2024, allowing for an exploration of both foundational theories and contemporary advancements in AI in education.
4. **Inclusion and Exclusion Criteria:** Studies were selected based on predefined inclusion and exclusion criteria to ensure relevance and quality. The criteria are detailed in the [Table 2](#).
5. **Categorization:** Selected articles were categorized according to the theoretical frameworks applied within the AI in education context, focusing on behaviorism, cognitivism, constructivism, and complexity theories. This categorization facilitated a deeper understanding of how different theories inform the development and implementation of AI technologies.
6. **Screening Process:** The screening process was conducted in multiple stages. Initially, two independent reviewers screened the titles and abstracts of all retrieved articles to identify potentially relevant studies. Articles that met the inclusion criteria during this initial screening were then subjected to a full-text review by the same reviewers. Any discrepancies between the reviewers were resolved through discussion or, when necessary, consultation with a third reviewer. To facilitate this process, the systematic review management software Covidence was employed, which aided in managing references and ensuring transparency and reproducibility (Moher et al., 2009). A PRISMA flow diagram illustrating the step-by-step process of study selection including the number of records identified, screened, assessed for eligibility, and included in the review is provided in [Figure 1](#).

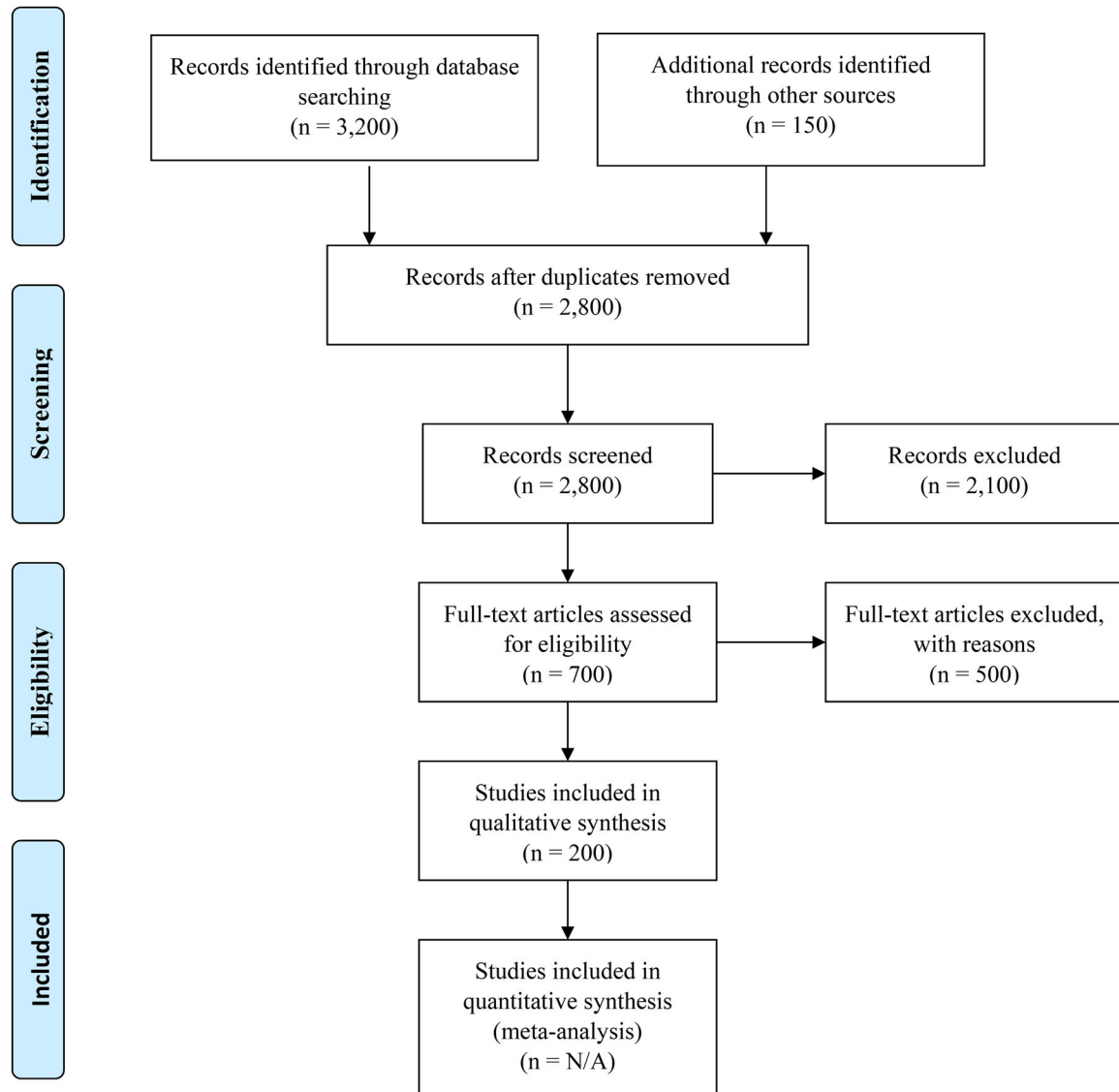
Table 1. Search strategy.

Concept	Keywords and search terms
Artificial Intelligence	"Artificial intelligence" OR AI OR "machine learning" OR "deep learning" OR "natural language processing" OR "intelligent tutor*" OR "adaptive system*"
Educational Theories	behaviorism OR cognitivism OR constructivism OR "learning theory" OR "instructional design" OR "pedagogical framework*"
Education Context	education* OR "learning environment*" OR "teaching practice*" OR "instructional method*" OR "educational technology*"
Combined Search String	("artificial intelligence" OR AI OR "machine learning" OR "deep learning" OR "natural language processing" OR "intelligent tutor*" OR "adaptive system*") AND (behaviorism OR cognitivism OR constructivism OR "learning theory" OR "instructional design" OR "pedagogical framework*") AND (education* OR "learning environment*" OR "teaching practice*" OR "instructional method*" OR "educational technology*")

The use of truncation symbols (e.g., *) allows for the inclusion of various word endings, enhancing the comprehensiveness of the search.

Table 2. Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Peer-reviewed journal articles and conference papers	Non-peer-reviewed articles, opinion pieces, editorials
Studies published between 2000 and 2024	Studies published before 2000 or after 2024
Articles written in English	Articles written in languages other than English
Studies focusing on the application of AI in educational settings	Studies focusing on AI applications outside the educational context
Research integrating AI with established educational theories	Studies lacking a connection to educational theories
Empirical studies, systematic reviews, and theoretical papers	Case studies without theoretical or empirical analysis

**Figure 1.** PRISMA flow diagram of study selection process.

7. **Data Extraction and Coding:** A standardized data extraction form was developed to collect pertinent information from each included study, such as publication details (author(s), year, journal/conference), study objectives and research questions, AI technologies and educational theories addressed, methodological approaches, and key findings and conclusions. The coding process involved thematic analysis to categorize studies according to the theoretical frameworks applied within the AI in education context, focusing on behaviorism, cognitivism, constructivism, and complexity theories. This categorization facilitated a deeper understanding of how different theories inform the development and implementation of AI technologies.
8. **Analysis:** The identified literature was analyzed to uncover the educational theories that underpin AI applications, the various roles AI plays in facilitating learning, and the broader implications of

these technologies for educational practices. This analysis emphasized critical insights into how AI can enhance or hinder the educational experience.

Results

The examination of AI in education reveals significant paradigmatic shifts, classified into three primary frameworks: AI-Guided Learning, AI-Interactive Learning, and AI-Driven Personalized Learning as shown in Table 3. Each framework reflects a distinct approach to integrating AI into educational practices, highlighting the evolving nature of learning environments.

Table 3 emphasizes the evolution of AIEd from structured instruction to collaborative exploration and ultimately to learner empowerment, showcasing the growing role of human-centered AI in education. It aligns with contemporary trends in personalized learning and ethical AI integration.

Paradigm one: AI-guided learning

The “AI-Guided Learning” paradigm emphasizes the use of AI systems that provide structured content and guide learners along predetermined pathways. This approach is rooted in behaviorism, which posits that learning occurs through reinforcement and feedback in response to stimuli (Greeno et al., 1996; Ouyang & Jiao, 2021; Skinner, 1953). Traditional ITSs, such as the ACT Programming Tutor, exemplify this paradigm by utilizing statistical methods to tailor instructional sequences based on learners’ inputs (Anderson et al., 1995). While this model has proven effective for knowledge acquisition, it often lacks flexibility in addressing individual learner needs, resulting in a rigid and sometimes disengaging learning experience (Du Boulay, 2019; Kay, 2000). While AI systems can adapt to learner inputs, they often fall short in addressing diverse learning needs and preferences. The reliance on static content delivery can lead to disengagement, particularly for learners who require more interactive or adaptive experiences. Traditional ITSs, like the ACT Programming Tutor, effectively use algorithm-driven assessments to evaluate learner performance and provide tailored feedback. However, such systems are criticized for lacking the flexibility to facilitate deeper engagement or foster critical thinking skills (Du Boulay, 2019; Kay,

Table 3. Evolving paradigms in AIEd.

Paradigm	Theory used	How it was used	AI techniques	Scholars and examples
Paradigm One: AI-Guided Learning	Rooted in behaviorist learning theories, focusing on reinforcing knowledge through repetition and immediate feedback. The paradigm views learners as passive recipients of knowledge.	Early AI-based systems delivering structured learning through predetermined tasks and feedback loops. These systems primarily emphasized skill acquisition and error correction.	Rule-based systems to deliver structured tasks; probabilistic analysis for learner modeling.	AI tools like the SOAR Cognitive Architecture (Laird, 2012) and early Drill-and-Practice Programs emphasized controlled learning experiences.
Paradigm Two: AI-Interactive Learning	Built on constructivist and socio-cognitive theories, emphasizing collaboration, interaction, and the co-construction of knowledge. Learners actively engage in the learning process.	Interactive systems such as dialogue-based learning environments and adaptive tutoring systems facilitate learner engagement, inquiry-based activities, and problem-solving.	Techniques include Bayesian inference models for predicting learner progress; Natural Language Processing (NLP) for conversational interfaces; decision-tree algorithms for adaptive pathways.	Platforms like ALEKS Adaptive Learning (Falmagne et al., 2013) and Woolf’s intelligent tutors created collaborative, exploratory settings.
Paradigm Three: AI-Driven Personalized Learning	Based on connectivism and complex adaptive systems, this paradigm empowers learners as active decision-makers and leaders of their educational journeys.	Advanced systems provide personalized learning and enable real-time collaboration between learners and intelligent agents. These systems prioritize adaptability and learner agency.	Leveraging Deep Learning for personalized feedback, Real-Time Predictive Models for dynamic content adjustment, and Brain-Computer Interfaces (BCIs) for understanding neural activity.	Innovations like Knewton Adaptive Learning (Baker & Inventado, 2014) and VR-AI educational platforms support active learner involvement and personalization.

2000). Educators must consider integrating more dynamic and interactive elements into AI systems to enhance engagement and motivation, moving beyond a purely directive approach to a more exploratory learning environment.

This paradigm emphasizes reinforcement through repetition and feedback, which has dominated early AIED systems such as ITSs. These systems provided structured guidance based on predefined rules and pathways, utilizing statistical relational techniques and rule-based systems to analyze learner input and provide targeted outputs. Examples include the ACT Programming Tutor and Stat Lady, which offer structured instruction and targeted exercises (Burns & Capps, 2013; Graesser et al., 2005; VanLehn et al., 2005). Overall, this paradigm focuses on linear, directive approaches to learning, with systems like the SOAR Cognitive Architecture exemplifying this model (Laird, 2012).

The AI-Guided Learning paradigm, rooted in behaviorism, employs AI systems like ITSs to guide learners along predetermined pathways, emphasizing repetition, reinforcement, and structured feedback (Anderson et al., 1995; Skinner, 1953). While effective for basic knowledge acquisition, this approach faces notable challenges. It lacks flexibility to accommodate diverse learner needs or adapt to unique preferences, often resulting in rigid, disengaging experiences (Du Boulay, 2019; Kay, 2000). Furthermore, its emphasis on static, algorithm-driven assessments restricts opportunities for fostering critical thinking and creativity (Graesser et al., 2005; VanLehn, 2005). These systems, such as the ACT Programming Tutor and early tools like PLATO, prioritize linear, directive methods that neglect socio-emotional and contextual factors critical for holistic learning (Burns & Capps, 2013; Laird, 2012; Ouyang & Jiao, 2021). Moreover, their reliance on behaviorist principles limits integration of more interactive or constructivist learning theories, underscoring the need for dynamic and engaging AI models in education (Greeno et al., 1996).

AI-guided learning systems, grounded in behaviorist theories, have been effectively utilized to provide structured and repetitive instruction. For instance, the AutoTutor system has demonstrated significant learning gains in subjects like computer literacy and conceptual physics, with effect sizes averaging 0.8, equivalent to a full letter grade improvement. Similarly, Squirrel AI Learning in China employs an Intelligent Adaptive Learning System (IALS) to tailor lesson plans to individual students by identifying specific knowledge gaps and providing targeted instruction, thereby enhancing learning efficiency.

Paradigm two: AI-interactive learning

The second paradigm, "AI-Interactive Learning," shifts the focus to collaborative engagement between learners and AI systems. Grounded in social constructivist theories, this approach emphasizes the importance of interactions among learners, peers, and technology (Bandura, 1986; Vygotsky, 1978). AI systems in this framework serve as facilitators, adapting to individual learners' needs by collecting and analyzing real-time data, thus creating personalized learning experiences (Baker et al., 2019; Rose et al., 2019). For example, dialogue-based tutoring systems allow learners to engage in meaningful interactions, prompting them to ask questions and explore the reasoning behind AI-generated responses (Metzler & Martincic, 1998). Despite the advancements in this area, challenges remain in effectively integrating learner data into AI systems, which is crucial for fostering continuous improvement and adaptability (UNESCO, 2019). Systems such as dialogue-based tutoring exemplify this by prompting learners to think critically and articulate their reasoning (Metzler & Martincic, 1998). The effectiveness of AI-Interactive Learning hinges on the ability of AI systems to analyze learner data dynamically. Successful implementations demonstrate improved learner outcomes by adapting to individual needs and providing personalized support (Baker et al., 2019; Ouyang & Jiao, 2021; Rose et al., 2019). Despite these advancements, challenges persist in effectively leveraging learner data for continuous system improvement. AI systems must ensure data privacy and ethical considerations while maintaining transparency in their operations (UNESCO, 2019).

This paradigm encourages active engagement and co-construction of knowledge among learners, facilitating interaction through adaptive dialogue systems and simulations. AI techniques such as Bayesian networks and Natural Language Processing (NLP) are employed to adapt content based on learner behavior and enable conversational interactions. Notable implementations include the QUE Exploratory Environment, which supports inquiry-based learning, and the ALEKS adaptive system, which

combines Bayesian inference with exploratory mathematics learning (Chi et al., 2018; Falmagne et al., 2013; Luckin et al., 2016; Ouyang & Jiao, 2021). This paradigm emphasizes the constructivist perspective, where learners are engaged in dialogue and scaffolded learning experiences.

The AI-Interactive Learning paradigm, grounded in social constructivist theories, emphasizes collaboration between learners and AI systems, with a focus on co-constructing knowledge through interaction and personalization (Bandura, 1986; Vygotsky, 1978). AI systems in this framework adaptively respond to learners by analyzing real-time data, enabling personalized pathways and fostering critical thinking through dialogue-based systems (Baker et al., 2019; Luckin et al., 2016). Tools like the QUE Exploratory Environment and ALEKS adaptive system exemplify this approach, offering inquiry-based and personalized learning experiences (Chi et al., 2018; Falmagne et al., 2013). However, several challenges persist, including the difficulty of dynamically integrating learner data for continuous improvement and adapting content to diverse learner needs (Ouyang & Jiao, 2021; Rose et al., 2019; UNESCO, 2019). Furthermore, ensuring ethical use of data, maintaining transparency, and addressing potential biases in AI algorithms are ongoing concerns (Holstein et al., 2018; Slade & Prinsloo, 2013). While these systems can enhance engagement and learning outcomes, they must balance adaptivity with robust ethical and privacy safeguards to fully realize their potential in educational contexts.

Interactive AI systems, influenced by constructivist theories, facilitate collaborative and personalized learning experiences. The ALEKS adaptive learning platform, for example, utilizes Bayesian inference to assess student knowledge and provide customized learning paths, resulting in improved engagement and mastery in mathematics. Additionally, the QUE Exploratory Environment supports inquiry-based learning by allowing students to interact with AI-driven simulations, promoting deeper understanding through exploration and dialogue.

Paradigm three: AI-driven personalized learning

The AI-Driven Personalized Learning paradigm redefines the learner's role, promoting agency and leadership in the learning process. This approach aligns with complexity theory, recognizing education as a dynamic, interconnected system (Mason, 2008). In this framework, AI serves as a tool to enhance human intelligence, promoting a collaborative ecosystem among learners, educators, and technology (Kovari, 2025; Ouyang & Jiao, 2021; Riedl, 2019). AI technologies empower learners to take charge of their educational journeys, encouraging self-directed exploration and critical engagement with content (Kovari, 2025; Riedl, 2019). Learners can leverage AI tools to analyze data, draw insights, and make informed decisions about their learning paths. The integration of AI fosters a collaborative ecosystem where learners, educators, and technology coalesce to create enriched learning experiences. Advanced techniques, such as machine learning, enable real-time adaptation of educational strategies to meet evolving learner needs (Hwang et al., 2020). While this paradigm offers significant potential for enhancing educational practices, it also raises concerns regarding the complexities of aligning AI systems with human values and educational goals (Knox, Williamson et al., 2020; Zawacki-Richter et al., 2019). Educators must critically assess how AI tools can support rather than hinder holistic learning experiences.

This paradigm prioritizes adaptability, personalization, and collaboration between learners and AI systems, allowing individuals to set goals and reflect on their learning journeys. Advanced AI techniques such as BCIs, machine learning, and real-time predictive modeling are utilized to provide dynamic, personalized learning experiences. Examples include predictive modeling in MOOCs and platforms like AltSchool, which create personalized learning paths for students (Baker & Inventado, 2014; Chandler, 2015; Liu et al., 2022). This paradigm reflects a shift towards human-centered AI, where learners engage dynamically with technology to enhance their educational experiences (Baxter & Hainey, 2024).

Human-computer cooperation systems play a vital role in the "AI-Driven Personalized Learning" paradigm by integrating advanced AI techniques with human decision-making to create adaptive, personalized, and interactive learning environments. These systems foster a dynamic partnership where AI supports learners in setting goals, making informed decisions, and reflecting on their progress, while human educators provide contextual, ethical, and emotional guidance. By utilizing machine learning, predictive modeling, and real-time analytics, AI tools adapt educational strategies to meet the diverse and evolving needs of learners (Hwang et al., 2020; Liu et al., 2022). For example, BCIs enable unique

accessibility solutions for students with disabilities, providing a new dimension of inclusion and personalization (Baxter & Hailey, 2024). In platforms like AltSchool, AI helps design personalized learning paths by analyzing learner data, while educators contribute their insights to ensure the alignment of AI recommendations with holistic educational goals (Chandler, 2015). MOOCs leverage predictive modeling to identify learners at risk of disengagement, empowering instructors to intervene effectively (Baker & Inventado, 2014; Liu et al., 2022). This cooperation system highlights the importance of balancing AI's computational efficiency with the human ability to foster creativity, critical thinking, and ethical decision-making. While AI augments learners' capabilities and adapts dynamically to their needs, educators ensure the integration of human values and long-term educational objectives, addressing concerns about over-reliance on algorithm-driven systems (Knox, Williamson et al., 2020; Zawacki-Richter et al., 2019). The collaborative ecosystem created by human-computer cooperation in this paradigm not only enriches learning experiences but also promotes the development of agency, leadership, and lifelong learning skills.

The paradigm faces several challenges, including concerns about data privacy and ethical issues surrounding the collection and use of learner data, which must balance security with transparency (Knox, Williamson et al., 2020; Zawacki-Richter et al., 2019). Bias in AI systems can perpetuate inequalities, while ensuring alignment with human values and educational goals remains a complex task, often risking a focus on efficiency at the expense of creativity and social-emotional learning (Baker & Inventado, 2014; Riedl, 2019). Additionally, over-reliance on AI could reduce the role of educators, risking dehumanization in education (UNESCO, 2019). Accessibility and implementation barriers, especially in low-resource contexts, highlight issues of cost, infrastructure, and training limitations (Hwang et al., 2020; Kovari, 2025; Kyambade & Namatovu, 2025). Moreover, designing AI systems that foster trust and meaningful human-AI collaboration is technically and pedagogically challenging, with scalability and adaptability to diverse educational needs adding further complexity (Baxter & Hailey, 2024; Liu et al., 2022). These challenges necessitate ethical AI practices, educator involvement, and equitable policies to ensure inclusive and effective learning experiences.

In the realm of personalized learning, AI technologies empower students to take control of their educational journeys. Platforms like Knewton's adaptive learning system have been shown to enhance student performance by providing real-time feedback and personalized content adjustments, leading to increased motivation and academic success. Furthermore, the integration of BCIs in educational settings offers unique accessibility solutions for students with disabilities, enabling personalized and inclusive learning experiences.

Figure 2 presents a conceptual framework outlining the three evolving paradigms of AI in education that's to say AI-Guided Learning, AI-Interactive Learning, and AI-Driven Personalized Learning. These paradigms represent distinct approaches to integrating AI into educational settings, shaped by different learning theories and technological advancements. **AI-Guided Learning**, rooted in behaviorist principles, focuses on structured instruction, reinforcement, and immediate feedback, utilizing rule-based algorithms and deterministic learning pathways. **AI-Interactive Learning**, grounded in constructivist and socio-cognitive theories, promotes learner engagement through adaptive tutoring, dialogue-based learning, and interactive problem-solving, with AI techniques such as Bayesian inference models and Natural Language Processing (NLP) enhancing dynamic learning experiences. **AI-Driven Personalized Learning**, influenced by connectivist and complex adaptive system theories, empowers learners as active decision-makers by leveraging deep learning, real-time predictive models, and BCIs to create highly personalized and adaptive learning environments. The framework underscores the shift from rigid, structured learning to interactive and ultimately personalized AI-driven educational experiences, emphasizing the growing role of human-centered AI in education.

Real life examples of AIED in Africa

Integrating AI into educational frameworks has the potential to revolutionize learning experiences across Africa, including Uganda. AI technologies can address challenges such as limited access to quality education, resource constraints, and language barriers. For instance, Makerere University in Uganda has developed the MusickingMaths App, an interactive mobile application that combines music with mathematics

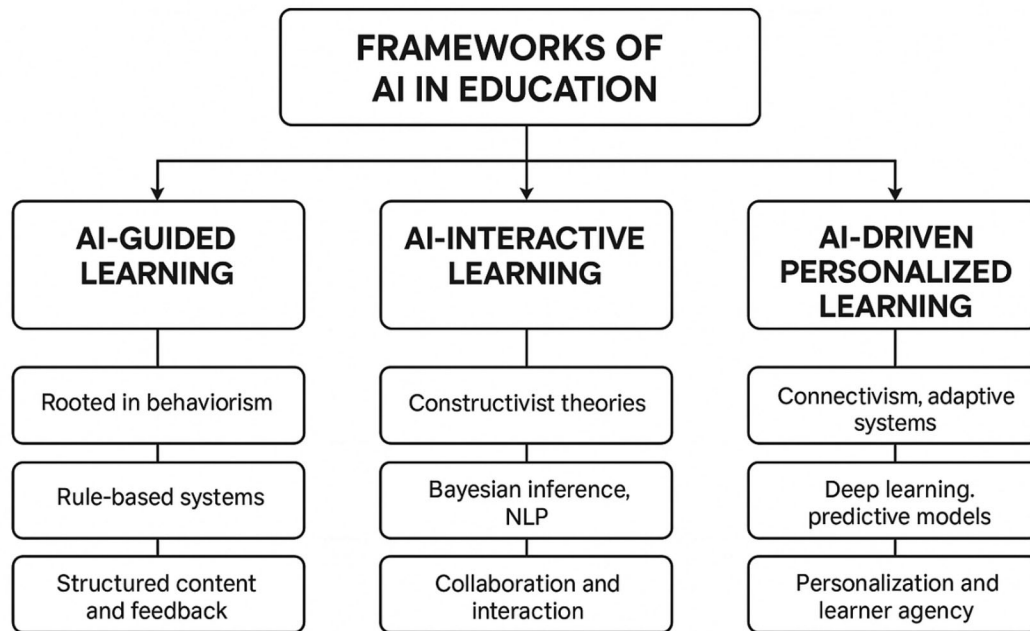


Figure 2. Evolving paradigms of AI in education: a three-tier framework.

to enhance learning among lower secondary students. This app allows students to create songs based on mathematical concepts, making learning more engaging and effective.

In West Africa, the AI teaching assistant “Kwame for Science” has demonstrated significant promise. Developed to assist students preparing for the West African Senior Secondary Certificate Examination, Kwame provides instant answers to science-related questions by delivering relevant passages and related past exam questions. A real-world deployment involving 750 users across 32 countries, including 15 African nations, reported an 87.2% top-three accuracy rate, highlighting its effectiveness in supporting science education.

Additionally, AI-driven platforms like M-Shule in Kenya offer personalized educational content and assessments for primary school students. Utilizing AI algorithms, M-Shule adapts to each student’s learning needs, providing tailored feedback to both students and teachers. Similarly, Uthini, an AI language learning app in South Africa, focuses on teaching users to speak and understand Xhosa. Employing speech recognition and machine learning, Uthini offers interactive language lessons with pronunciation feedback, demonstrating AI’s capacity to bridge linguistic gaps in education.

These examples underscore AI’s transformative role in African education, offering personalized learning experiences, supporting educators, and preserving linguistic diversity. As AI technologies continue to evolve, their integration into educational systems across Africa, including Uganda, holds promise for more inclusive, accessible, and effective learning environments.

Discussion

Educational and learning theories have often been overlooked in the design of AIED systems. Scholars argue for a closer integration of AI technologies with established educational and learning theories to create more effective and meaningful applications (Chen et al., 2020; Hopcan et al., 2023; Hwang et al., 2020; Hwang & Tu, 2021). This study identifies three paradigms in AIED development: AI-Guided Learning, AI-Interactive Learning, and AI-Driven Personalized Learning. While the first two paradigms have dominated AIED research and practice over the past three decades (Ouyang & Jiao, 2021; Wang et al., 2024), the emerging third paradigm represents a critical shift towards fostering learner agency and empowerment.

The AI-Driven Personalized Learning paradigm emphasizes integrating human intelligence with AI systems to overcome challenges such as algorithmic bias, governance issues, and lack of transparency (Hwang & Tu, 2021; Lee & Kwon, 2024; Salas-Pilco & Yang, 2022). This approach calls for the adoption of advanced multimodal data collection, real-time AI models, and a focus on multidimensional human attributes including cognitive, social, emotional, and ethical factors to create personalized and learner-centered experiences.

1. **Multimodal Data Collection**—Innovations in data collection, such as physiological sensors, eye-tracking, and electroencephalography, enable a comprehensive understanding of learners' behaviors, emotions, and cognitive states. Studies highlight how these techniques improve learning performance predictions and offer valuable insights into the dynamics of human learning (Cukurova et al., 2019; Giannakos et al., 2019; Xie et al., 2019). As multimodal systems evolve, they can support richer interpretations of learner needs and promote adaptive learning systems.
2. **Real-Time AI Algorithms**—Real-time AI systems provide instantaneous feedback to learners and educators, supporting dynamic learning interactions (Zhang & Tur, 2024). The integration of real-time data streams with human-computer interaction (HCI) approaches facilitates personalized educational strategies. For instance, AI-driven dashboards can enable educators to track learner progress and make informed instructional decisions (Dogan et al., 2023; Giannakos et al., 2019). However, effective implementation requires balancing technological capabilities with ethical considerations, such as data privacy and transparency (Kovari, 2025; Kyambade & Namatovu, 2025).
3. **Human-Centered AI (HAI)**—Moving towards a human-centered paradigm involves designing AI systems that account for the unique attributes of human learners such as emotions, reflective judgment, and creative thinking that AI cannot replicate. By aligning AI capabilities with educational goals, HAI emphasizes empowering learners to take ownership of their learning journeys (Kovari, 2025; Lin et al., 2023). This paradigm requires embedding reflective practices and providing learners with the tools to understand and influence how AI systems process and use their data (Verkijika & De Wet, 2015).

While advanced technologies, such as BDIs, offer significant potential, they often fail to fully align with learner-centric principles. Many applications prioritize technical precision over the learner's agency and understanding of their role in the educational process. For AIED to reach its full potential, developers must design systems that integrate multimodal data, human cognition, and ethical AI principles (Labadze et al., 2023). Researchers highlight the importance of fostering partnerships between AI systems and learners, creating iterative feedback loops that enable both to grow and adapt dynamically (Cukurova et al., 2019; Knox, Williamson et al., 2020; Ouyang & Jiao, 2021).

The future of AIED lies in transitioning from directive and collaborative paradigms to an empowered paradigm, where learners are not passive recipients of AI-driven content but active participants in their educational journeys (Mohebbi, 2025). This shift requires designing systems that balance technological innovation with human-centric values, creating environments that empower learners to reflect, adapt, and lead. By prioritizing human-AI collaboration, education can foster deeper engagement, critical thinking, and lifelong learning.

The examination of AI in education reveals a significant evolution in pedagogical paradigms, transitioning from AI-Guided Learning (Ouyang & Jiao, 2021), where learners are seen as passive recipients, to AI-Driven Personalized leadership, where learners actively shape their educational journeys (Ouyang & Jiao, 2021; Salas-Pilco & Yang, 2022). This transition highlights a growing recognition of the learner's role as an agent of change, fostering agency, leadership, and collaboration in their learning processes. In the AI-Guided Learning paradigm, rooted in behaviorism, early AI systems provided structured, feedback-driven learning experiences that were effective for knowledge acquisition but often fell short of engaging diverse learner needs and promoting critical thinking (Du Boulay, 2019; Kay, 2000; Skinner, 1953). These systems, while beneficial for basic skills development, exemplified a rigid approach that limited learner engagement and exploration.

In contrast, the AI-Interactive Learning paradigm emphasizes interactive learning experiences grounded in social constructivist theories, where learners actively participate in co-constructing

knowledge alongside AI systems (Bandura, 1986; Vygotsky, 1978). This shift towards collaborative frameworks is facilitated by advanced AI techniques, such as Natural Language Processing and adaptive tutoring systems, which personalize learning pathways and foster meaningful interactions (Baker et al., 2019; Rose et al., 2019). However, challenges remain, including the effective integration of learner data for continuous improvement and the ethical implications of data usage, which necessitate careful consideration by educators (UNESCO, 2019).

The AI-Driven Personalized Learning paradigm further redefines the learner's role by integrating advanced technologies, such as machine learning and BCIs, which empower learners to take charge of their educational journeys and promote self-directed exploration (Feigerlova et al., 2025; Hwang et al., 2020; Riedl, 2019). This dynamic partnership between AI and human decision-making fosters a collaborative ecosystem that enriches learning experiences and enhances adaptability to diverse learner needs. Nonetheless, the paradigm raises concerns about over-reliance on AI, which could diminish the role of educators and lead to dehumanization in educational contexts (Knox, Williamson et al., 2020; UNESCO, 2019).

Overall, while the integration of AI in education offers significant potential to transform learning environments, it also presents a complex array of challenges that necessitate ongoing dialogue among educators, policymakers, and technologists. Addressing issues of data privacy, algorithmic bias, and ethical practices will be crucial to ensuring that AI tools support holistic learning experiences rather than hinder them (Baker & Inventado, 2014; Zawacki-Richter et al., 2019). As education continues to evolve in the context of AI advancements, fostering trust, ensuring inclusivity, and promoting lifelong learning skills will be paramount in shaping the future of educational practices.

Conclusion

This article delineates three paradigms of AIEd that's AI-Guided Learning, AI-Interactive Learning, and AI-Driven Personalized Learning each reflecting a progressive shift toward more personalized and learner-centered educational experiences. These paradigms illustrate the evolving role of AI from directing instruction to supporting collaboration and ultimately empowering learners to take charge of their educational journeys. The integration of AI into education necessitates a holistic approach that combines technological innovation with pedagogical frameworks and considers social, cultural, and economic contexts to ensure effective adoption across diverse learning environments (Castañeda & Selwyn, 2018; Selwyn, 2016).

Future advancements in AIEd should prioritize learner-centered design principles, emphasizing data-driven personalization and adaptability to meet the needs of modern learners (Deeva et al., 2021; Holmes et al., 2019). This includes embedding ethical, cognitive, and cultural dimensions within AI systems to foster equitable, inclusive, and empowering learning environments (Knox, Wang et al., 2020; Kovari, 2025). The evolution of AIEd must transcend technological innovation alone, addressing the complexities of educational ecosystems by incorporating a balanced approach to human-centered AI design and educational theory. Such an approach aims to build adaptive, impactful, and meaningful learning experiences for all stakeholders (Luckin et al., 2016; Zawacki-Richter et al., 2019).

Recommendations

To effectively harness the potential of AI in African education, it is essential to implement strategies tailored to the continent's unique challenges and opportunities. The following recommendations are proposed: Firstly, Establish ethical frameworks to govern data usage, privacy, and AI applications, ensuring alignment with African cultural and societal values. This approach fosters trust and promotes the responsible use of AI technologies in educational contexts. Secondly, Adopt AI systems that provide transparent decision-making processes, enabling educators and students to understand and trust AI-driven outcomes. Explainable AI enhances the credibility of AI tools and supports informed decision-making in educational settings. Thirdly, provide ongoing training for educators to effectively integrate AI tools into teaching practices, enhancing their digital literacy and pedagogical skills. Initiatives like the African Centre of Excellence in Data Science (ACE-DS) at the University of Rwanda exemplify such efforts,

offering certification programs in data science and AI to equip educators with necessary competencies. Address the digital divide by ensuring equitable access to AI technologies across urban and rural areas, fostering inclusive educational opportunities. Bridging this gap is crucial for enabling all students to benefit from AI-enhanced learning experiences. Encourage the development of AI solutions tailored to African educational contexts. For example, “Kwame for Science” is an AI teaching assistant designed to assist students by answering questions based on the Integrated Science curriculum of the West African Senior Secondary Certificate Examination (WASSCE). Deployed as a web app, it provides detailed answers along with related past exam questions, demonstrating the potential of localized AI solutions to enhance science education across multiple African countries. Promote partnerships between African educational institutions and global AI research centers to develop contextually relevant AI applications. The establishment of ACE-DS, a World Bank-funded center of excellence in data science, exemplifies such collaborative efforts, aiming to train postgraduate students and build research capacity in data science to address regional challenges. Utilize AI to develop language models for African languages, preserving linguistic diversity and enhancing educational content accessibility. Collaborations like those between Orange, Meta, and OpenAI aim to refine AI language models for African languages, supporting the development of educational resources that reflect the continent’s rich linguistic heritage.

Limitations and future research directions

This systematic literature review on AIED presents several limitations that should be acknowledged. Firstly, the review was confined to peer-reviewed articles published in English between 2000 and 2024. This language restriction may have excluded relevant studies published in other languages, potentially limiting the comprehensiveness of the review. Secondly, while efforts were made to include a broad range of databases such as Web of Science, Scopus, and IEEE, the exclusion of grey literature and non-indexed sources might have led to publication bias. This bias arises when studies with non-significant or negative results are less likely to be published, thereby skewing the overall findings. Thirdly, the heterogeneity of the included studies in terms of methodologies, educational contexts, and AI applications posed challenges in synthesizing the findings. This diversity limited the ability to perform a meta-analysis and may affect the generalizability of the conclusions drawn. Lastly, the rapidly evolving nature of AI technologies means that new developments may have emerged since the completion of this review. Consequently, the findings may not fully capture the most recent advancements in AIED. Future research should consider including non-English publications, grey literature, and employ methodologies that can accommodate the heterogeneity of studies to provide a more comprehensive understanding of AI applications in education.

Future research should focus on: Conducting empirical studies to assess the effectiveness of AI applications in improving learning outcomes within African educational settings. Such evaluations are essential for understanding the tangible benefits and areas for improvement in AI integration. Creating AI-driven educational materials that reflect African cultural contexts and languages, enhancing relevance and engagement. Developing content that resonates with students’ cultural backgrounds can improve learning experiences and outcomes. Investigating solutions to infrastructural barriers hindering AI adoption, such as internet connectivity and electricity reliability. Overcoming these challenges is vital for ensuring the seamless integration of AI technologies in educational environments. Exploring frameworks to prevent biases in AI algorithms and protect user privacy, aligning with ethical standards and societal norms. Establishing robust ethical guidelines is crucial for maintaining trust and promoting the responsible use of AI in education. By implementing these recommendations and pursuing targeted research, AI can significantly contribute to transforming education across Africa, making it more personalized, inclusive, and aligned with the continent’s developmental goals.

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References

- Al Mansoor, A. (2023). Investing in professional development for AI integration in education. *Journal of Educational Technology Development and Exchange*, 16(2), 45–58.
- Alfredo, R., Echeverria, V., Jin, Y., Yan, L., Swiecki, Z., Gašević, D., & Martinez-Maldonado, R. (2024). Human-centred learning analytics and AI in education: A systematic literature review. *Computers and Education: Artificial Intelligence*, 100215.
- Altukhi, Z. M., & Pradhan, S. (2025). Systematic literature review: Explainable AI definitions and challenges in education *arXiv preprint arXiv:2504.02910* [https://arxiv.org/abs/2504.02910​;contentReference\[oaicite:26\]{index = 26}](https://arxiv.org/abs/2504.02910​;contentReference[oaicite:26]{index = 26})
- Anderson, J. R., Corbett, A. T., Koedinger, K. R., & Pelletier, R. (1995). Cognitive tutors: Lessons learned. *Journal of the Learning Sciences*, 4(2), 167–207. https://doi.org/10.1207/s15327809jls0402_2
- Baker, R. S. J. D., & Inventado, P. S. (2014). Educational data mining and learning analytics. In *Learning, design, and technology: An international compendium of theory, research, practice, and policy* (pp. 1–23). Springer.
- Baker, R. S., & Inventado, P. S. (2019). Educational data mining and learning analytics. In *Handbook of technology and second language teaching and learning* (pp. 72–83).
- Baker, R. S., Inventado, P. S., & Gowda, S. M. (2019). Educational data mining and learning analytics in intelligent tutoring systems. In R. Nkambou, R. Mizoguchi, & J. Bourdeau (Eds.), *Advances in intelligent tutoring systems* (pp. 61–75). Springer.
- Bandura, A. (1986). *Social foundations of thought and action: A social cognitive theory*. Prentice-Hall.
- Bates, J. (2023). *The future of AI in education: Opportunities and concerns*. Science Matterz.
- Baxter, G., & Hainey, T. (2024). Using immersive technologies to enhance the student learning experience. *Interactive Technology and Smart Education*, 21(3), 403–425. <https://doi.org/10.1108/ITSE-05-2023-0078>
- Bigscal Technologies. (2024). Benefits of artificial intelligence in education Retrieved from <https://medium.com/@Bigscal-Technologies/top-benefits-of-artificial-intelligence-in-education-sector-fa68b9e5f749>
- Bower, M. (2021). Educational design for AI: Principles, processes, and potential. *Computers in Human Behavior*, 116, 106635.
- Burns, H. L., & Capps, C. G. (2013). Foundations of intelligent tutoring systems: An introduction. In *Foundations of intelligent tutoring systems* (pp. 1–19). Psychology Press.
- Castañeda, L., & Selwyn, N. (2018). More than tools? Making sense of the ongoing digitizations of higher education. *International Journal of Educational Technology in Higher Education*, 15(1), 22. <https://doi.org/10.1186/s41239-018-0109-y>
- Chandler, D. (2015). *The Future of Personalized Learning: How AltSchool Is Reimagining K-12 Education*. EdSurge News.
- Chen, X., Xie, H., & Hwang, G. J. (2020). Trends and research issues of artificial intelligence in education: A review of journal publications from 1993 to 2020. *International Journal of Artificial Intelligence in Education*, 30, 1–32. <https://doi.org/10.1007/s40593-020-00209-2>
- Chen, X., Zou, D., Xie, H., & Hwang, G.-J. (2020). Application and theory gaps during the past three decades of AI in education. *Computers & Education*, 146, 103719.

- Chi, M., VanLehn, K., Litman, D., & Jordan, P. (2018). An evaluation of spoken dialogue systems for tutorial applications. *Computers and Education*, 61, 207–218.
- Cukurova, M., Luckin, R., & Clark-Wilson, A. (2019). Creating the golden triangle of evidence-informed education with edtech. *British Journal of Educational Technology*, 50(2), 490–504. <https://doi.org/10.1111/bjet.12727>
- Deeva, M., Holmes, W., & Porayska-Pomsta, K. (2021). Learner-centered artificial intelligence in education: The road ahead. *British Journal of Educational Technology*, 52(4), 1287–1299. <https://doi.org/10.1111/bjet.13135>
- Dogan, M. E., Goru Dogan, T., & Bozkurt, A. (2023). The use of artificial intelligence (AI) in online learning and distance education processes: A systematic review of empirical studies. *Applied Sciences*, 13(5), 3056. <https://doi.org/10.3390/app13053056>
- Du Boulay, B. (2019). AI as an existential threat to education. *British Journal of Educational Technology*, 50(3), 887–897.
- Du Boulay, B. (2021). Reflections on Artificial Intelligence in Education: From My Perspective. *International Journal of Artificial Intelligence in Education*, 31(3), 1–16.
- European Commission. (2023). Ethical guidelines on the use of artificial intelligence (AI) and data in teaching and learning for educators <https://op.europa.eu/en/publication-detail/-/publication/d81a0d54-5348-11ed-92ed-01aa75ed71a1/language-en>
- Falmagne, J.-C., Albert, D., Doble, C., Eppstein, D., & Hu, X. (2013). ALEKS: A Web-based intelligent tutoring system. *International Journal of Artificial Intelligence in Education*, 13(2), 235–265.
- Feigerlova, E., Hani, H., & Hothersall-Davies, E. (2025). A systematic review of the impact of artificial intelligence on educational outcomes in health professions education. *BMC Medical Education*, 25(1), 129. <https://doi.org/10.1186/s12909-025-06719-5>
- Forero-Corba, W., & Bennasar, F. N. (2023). Técnicas y aplicaciones del machine learning e inteligencia artificial en educación: Una revisión sistemática. *RIED-Revista Iberoamericana de Educación a Distancia*, 27(1), 209–253. <https://doi.org/10.5944/ried.27.1.37491>
- Giannakos, M. N., Sharma, K., & Pappas, I. O. (2019). Exploring multimodal data in learning: Towards an interactional understanding of learning. *British Journal of Educational Technology*, 50(6), 2767–2783.
- Gillani, N., Eynon, R., Chiabaut, C., & Finkel, K. (2023). Unpacking the “Black Box” of AI in Education. *Educational Technology & Society*, 26(1), 99–111. <https://www.jstor.org/stable/48707970>
- Graesser, A. C., Chipman, P., & King, B. (2005). Learning environments with conversational agents. *AI & Society*, 19(1), 1–8.
- Greeno, J. G., Collins, A., & Resnick, L. (1996). Cognition and learning. In D. C. Berliner, & R. C. Calfee (Eds.), *Handbook of educational psychology* (pp. 15–46). Macmillan.
- Hardaker, G., & Glenn, L. E. (2025). Artificial intelligence for personalized learning: A systematic literature review. *The International Journal of Information and Learning Technology*, 42(1), 1–14. <https://doi.org/10.1108/IJILT-07-2024-0160>
- Holmes, W., Bialik, M., & Fadel, C. (2019). Artificial intelligence in education: Promises and implications for teaching and learning Center for Curriculum Redesign. <https://curriculumredesign.org/>
- Holstein, K., Wortman, J., & Aleven, V. (2020). The role of teacher trust in AI integration. *International Journal of Artificial Intelligence in Education*, 30(4), 465–489. <https://doi.org/10.1007/s40593-020-00209-x>
- Holstein, K., Wortman, J., Aleven, V., & Klemmer, S. R. (2018). The ethics of adaptive learning systems. In *Proceedings of the learning analytics and knowledge conference (LAK)* (pp. 282–291). ACM.
- Hopcan, S., Polat, E., Ozturk, M. E., & Ozturk, L. (2023). Artificial intelligence in special education: A systematic review. *Interactive Learning Environments*, 31(10), 7335–7353. <https://doi.org/10.1080/10494820.2022.2067186>
- Hwang, G.-J., & Tu, Y.-F. (2021). Roles and research trends of artificial intelligence in STEM education: A review of journal publications from 1998 to 2018. *Computers & Education*, 151, 103857.
- Hwang, G. J., Xie, H., Wah, B. W., & Gašević, D. (2020). Vision, challenges, roles and research issues of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 1, 100001. <https://doi.org/10.1016/j.caeai.2020.100001>
- Informal Pakistan. (2024). Transforming Education: The Benefits of Artificial Intelligence in the Classroom Retrieved from <https://www.informal.pk/transforming-education-the-benefits-of-artificial-intelligence-in-the-classroom>
- Kay, J. (2000). Learner control: A framework for the use of learning technologies. *International Journal of Artificial Intelligence in Education*, 11(2), 164–195.
- Khan, W. N. (2025). Ethical challenges of AI in education: Balancing innovation with data privacy. *Journal of AI in Education: Innovations, Opportunities, Challenges, and Future Directions*, 1(1), 1–25. <https://www.researchcorridor.org/index.php/jaieicfd/article/view/238>
- Kim, K., & Kwon, K. (2025). A systematic review of the evaluation in K-12 artificial intelligence education from 2013 to 2022. *Interactive Learning Environments*, 33(1), 103–131. <https://doi.org/10.1080/10494820.2024.2335499>
- Knox, J., Wang, X., & Sun, Q. (2020). Critical perspectives on AI in education: Ethical and cultural dimensions. *AI & Society*, 35, 867–878. <https://doi.org/10.1007/s00146-020-01086-w>
- Knox, J., Williamson, B., & Bayne, S. (2020). Machine behaviourism: Future visions of “learnification” and “datafication” in education. *Learning, Media and Technology*, 45(1), 31–45. <https://doi.org/10.1080/17439884.2019.1623251>

- Kovari, A. (2025). A systematic review of AI-powered collaborative learning in higher education: Trends and outcomes from the last decade. *Social Sciences & Humanities Open*, 11, 101335. <https://doi.org/10.1016/j.ssaho.2025.101335>
- Kyambade, M., Abdul, S. M., & Namatovu, A. (2023). Adoption theories for internet banking in Uganda. *Advances in Research*, 24(6), 136–145. <https://doi.org/10.9734/air/2023/v24i6994>
- Kyambade, M., Kagere, R., Namatovu, A., Ssentumbwe, A. M., Tushabe, M., & Oyella, S. (2024). Technological innovation and commercial banks' financial performance: A mediation analysis of risk management practices. *SEISENSE Business Review*, 4(1), 98–117. <https://doi.org/10.33215/bb6wrs68>
- Kyambade, M., & Namatovu, A. (2025). Healthcare leaders' perspectives on AI implementation in Uganda: Overcoming barriers, driving innovation, and strategic considerations. *Leadership in Health Services*. <https://doi.org/10.1108/LHS-02-2025-0025>
- Kyambade, M., Namatovu, A., Ssentumbwe, A. M., & Tushabe, M. (2025). Work from home can be distracting: Exploring the moderation role of transformational leadership on teleworking and cyber-slacking. *Cogent Education*, 12(1), 2450115. <https://doi.org/10.1080/2331186X.2025.2450115>
- Kyambade, M., Tushabe, M., Mugerwa, G. W., & Namatovu, A. (2025). Am not contented at work: Unraveling socially responsible leadership and psychological safety on job satisfaction in Ugandan public universities. *Management Research Review*, 48(6), 863–882. <https://doi.org/10.1108/MRR-05-2024-0344>
- Labadze, L., Grigolia, M., & Machaidze, L. (2023). Role of AI chatbots in education: Systematic literature review. *International Journal of Educational Technology in Higher Education*, 20(1), 56. <https://doi.org/10.1186/s41239-023-00426-1>
- Laird, J. E. (2012). The SOAR cognitive architecture. *Cognitive Systems Research*, 10(2), 218–224.
- Lee, S. J., & Kwon, K. (2024). A systematic review of AI education in K-12 classrooms from 2018 to 2023: Topics, strategies, and learning outcomes. *Computers and Education: Artificial Intelligence*, 6, 100211.
- Lin, C. C., Huang, A. Y., & Lu, O. H. (2023). Artificial intelligence in intelligent tutoring systems toward sustainable education: A systematic review. *Smart Learning Environments*, 10(1), 41. <https://doi.org/10.1186/s40561-023-00260-y>
- Liu, S. (2024). *The future of AI in education: Opportunities and concerns*. Science Matterz.
- Liu, V., Latif, E., & Zhai, X. (2025). Advancing education through tutoring systems: A systematic literature review. arXiv Preprint, *arXiv:2503.09748*
- Liu, S., Liu, S., Liu, Z., Peng, X., & Yang, Z. (2022). Automated detection of emotional and cognitive engagement in MOOC discussions to predict learning achievement. *Computers & Education*, 181, 104461. <https://doi.org/10.1016/j.compedu.2022.104461>
- Lu, H. (2025). *AI doesn't shortchange learning. It enhances it*. San Francisco Chronicle.
- Luckin, R., Holmes, W., Griffiths, M., & Forcier, L. B. (2016). *Intelligence unleashed: An argument for AI in education*. Pearson Education.
- Maity, S., & Deroy, A. (2024). Generative AI and its impact on personalized intelligent tutoring systems. *arXiv preprint arXiv:2410.10650*
- Mason, J. (2008). Complexity theory and education. *Educational Studies in Mathematics*, 68(1), 89–103.
- Medium. (2024). Benefits of Artificial Intelligence in Education Retrieved from <https://medium.com/@Bigscal-Technologies/top-benefits-of-artificial-intelligence-in-education-sector-fa68b9e5f749>
- Metzler, T., & Martincic, P. (1998). Advances in dialogue-based intelligent tutoring systems. *International Journal of Artificial Intelligence in Education*, 9(3–4), 235–252.
- Mohebbi, A. (2025). Enabling learner independence and self-regulation in language education using AI tools: A systematic review. *Cogent Education*, 12(1), 2433814. <https://doi.org/10.1080/2331186X.2024.2433814>
- Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *BMJ (Clinical Research ed.)*, 339, b2535–b2535. <https://doi.org/10.1136/bmj.b2535>
- Namatovu, A., & Kyambade, M. (2025). Leveraging AI in academia: University students' adoption of ChatGPT for writing coursework (take home) assignments through the lens of UTAUT2. *Cogent Education*, 12(1), 2485522. <https://doi.org/10.1080/2331186X.2025.2485522>
- Novianti, I. (2025). An exploratory study on the challenges of AI technology in education and its practical recommendations. *International Journal of Social Science Research and Review*, 8(3), 26–37. <https://doi.org/10.47814/ijssrr.v8i3.2481>
- Ouyang, F., & Jiao, P. (2021). Artificial intelligence in education: The three paradigms. *Computers and Education: Artificial Intelligence*, 2, 100020.
- Ouyang, F., Zhu, S., Chen, B., & Dwivedi, Y. K. (2022). Artificial intelligence in education: Promises and pitfalls. *Computers & Education*, 180, Article104424. <https://doi.org/10.1016/j.compedu.2022.104424>
- Pinkwart, N. (2016). Another 25 Years of AIED? Challenges and Opportunities for Intelligent Educational Technologies of the Future. *International Journal of Artificial Intelligence in Education*, 26(2), 771–783. <https://doi.org/10.1007/s40593-016-0099-7>
- Porayska-Pomsta, K., Holmes, W., & Nemorin, S. (2023). The ethics of AI in education. In *Handbook of artificial intelligence in education* (pp. 571–604). Edward Elgar Publishing.

- Qin, Q., & Zhang, S. (2025). Visualizing the knowledge mapping of artificial intelligence in education: A systematic review. *Education and Information Technologies*, 30(1), 449–483. <https://doi.org/10.1007/s10639-024-13076-1>
- Riedl, M. O. (2019). Human-centered artificial intelligence and ethics in education. *IEEE Transactions on Learning Technologies*, 12(4), 475–483.
- Rivera, A. (2024). *The future of AI in education: Opportunities and Concerns*. Science Matterz.
- Rose, C. P., McLaughlin, E., Liu, R., & Koedinger, K. R. (2019). towards scalable personalized support for exploratory learning. *Journal of Learning Analytics*, 6(3), 1–16.
- Rowe, E., & Baepler, P. (2021). Learning analytics and artificial intelligence in higher education: Examining new opportunities. *Innovative Higher Education*, 46(4), 321–334.
- Salas-Pilco, S. Z., & Yang, Y. (2022). Artificial intelligence applications in Latin American higher education: A systematic review. *International Journal of Educational Technology in Higher Education*, 19(1) Article 21. <https://doi.org/10.1186/s41239-022-00326-w>
- Selwyn, N. (2016). Minding our language: Why education and technology is full of bullshit...and what might be done about it. *Learning, Media and Technology*, 41(3), 437–443. <https://doi.org/10.1080/17439884.2015.1012523>
- Skinner, B. F. (1953). *Science and human behavior*. Macmillan.
- Slade, S., & Prinsloo, P. (2013). Learning analytics: Ethical issues and dilemmas. *American Behavioral Scientist*, 57(10), 1510–1529. <https://doi.org/10.1177/0002764213479366>
- TeachEduicator. (2024). The role of artificial intelligence in education. Retrieved from <https://teacheducator.com/role-of-artificial-intelligence-in-education/Teach Educator>
- Tech AI. (2024). Advantages of artificial intelligence in education. Retrieved from <https://techai.tech/advantages-of-artificial-intelligence-in-education/>
- UNESCO. (2019). *Artificial intelligence in education: Challenges and opportunities for sustainable development*. UNESCO Publishing.
- UNESCO. (2023). The challenges and opportunities of Artificial Intelligence in education <https://www.unesco.org/en/articles/challenges-and-opportunities-artificial-intelligence-education>
- VanLehn, K. (2005). The behavior of tutoring systems. *International Journal of Artificial Intelligence in Education*, 16(3), 227–265.
- VanLehn, K., Siler, S., Murray, C., & Yamauchi, A. (2005). Why do only some students succeed in problem solving? *The Journal of the Learning Sciences*, 14(3), 261–298.
- Verkijika, S. F., & De Wet, L. (2015). Using a brain-computer interface to enhance attention in e-learning environments. *Computers & Education*, 86, 29–41.
- Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.
- Wang, S., Wang, F., Zhu, Z., Wang, J., Tran, T., & Du, Z. (2024). Artificial intelligence in education: A systematic literature review. *Expert Systems with Applications*, 252, 124167. <https://doi.org/10.1016/j.eswa.2024.124167>
- Ward, B., Bhati, D., Neha, F., & Guercio, A. (2024). Analyzing the impact of AI tools on student study habits and academic performance. arXiv preprint arXiv:2412.02166.
- White, E. (2024). *The future of AI in education: Opportunities and concerns*. Science Matterz.
- Wikipedia. (2024). Adaptive learning Retrieved from https://en.wikipedia.org/wiki/Adaptive_learning
- Xie, H., Jiang, C., & Zhang, L. (2019). Interactive AI-driven systems in education: Enhancing real-time feedback and learning performance. *International Journal of Educational Technology in Higher Education*, 16(1), 1–17.
- Yan, L., Sha, L., Zhao, L., Li, Y., Martinez-Maldonado, R., Chen, G., Li, X., Jin, Y., & Gašević, D. (2024). Practical and ethical challenges of large language models in education: A systematic scoping review. *British Journal of Educational Technology*, 55(1), 90–112. <https://doi.org/10.1111/bjet.13370>
- Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education – where are the educators? *International Journal of Educational Technology in Higher Education*, 16(1), 39. <https://doi.org/10.1186/s41239-019-0171-0>
- Zhang, C. (2024). AI in education: Opportunities, challenges, and pathways for equitable learning. *Journal of Education, Humanities and Social Sciences*, 45, 723–728. <https://doi.org/10.54097/kfgp6j07>
- Zhang, P., & Tur, G. (2024). A systematic review of ChatGPT use in K-12 education. *European Journal of Education*, 59(2), e12599. <https://doi.org/10.1111/ejed.12599>